

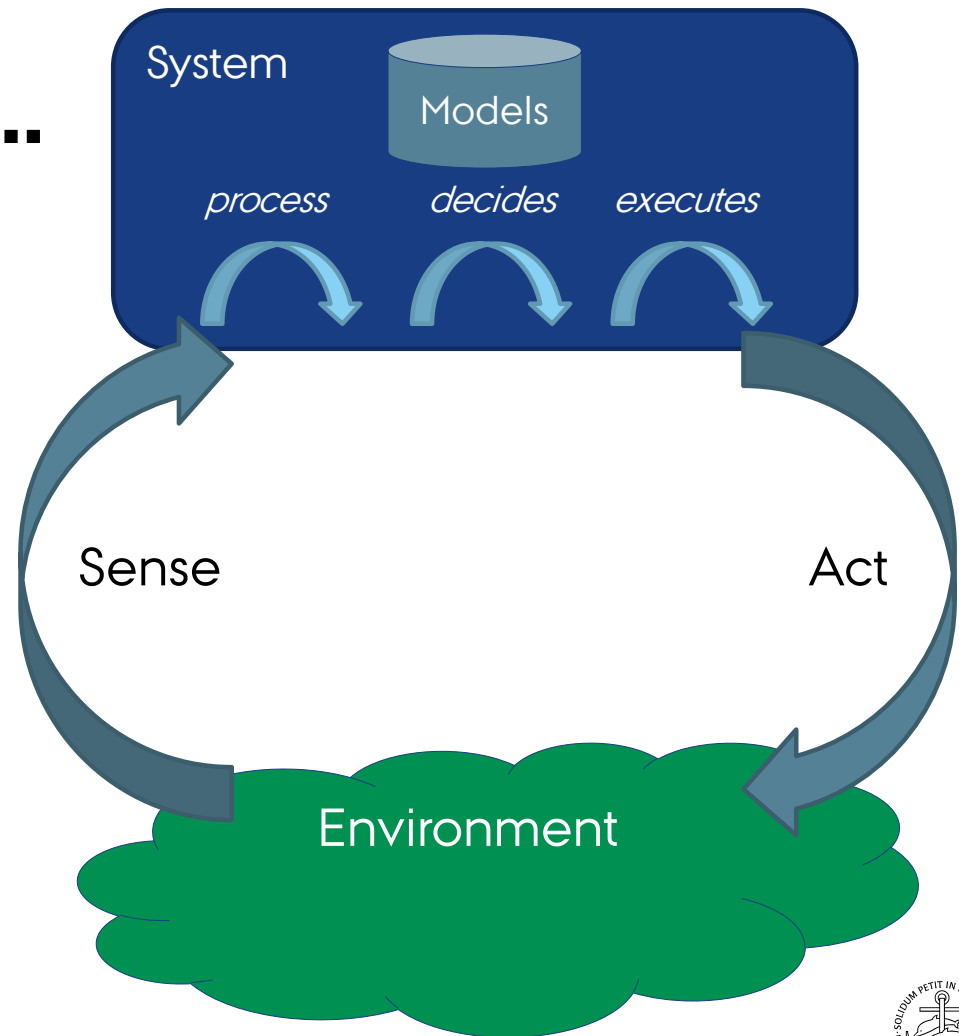
AUTONOMOUS AND MULTI-AGENT SYSTEMS NATURE-INSPIRED APPROACHES AND EMERGENT BEHAVIOUR

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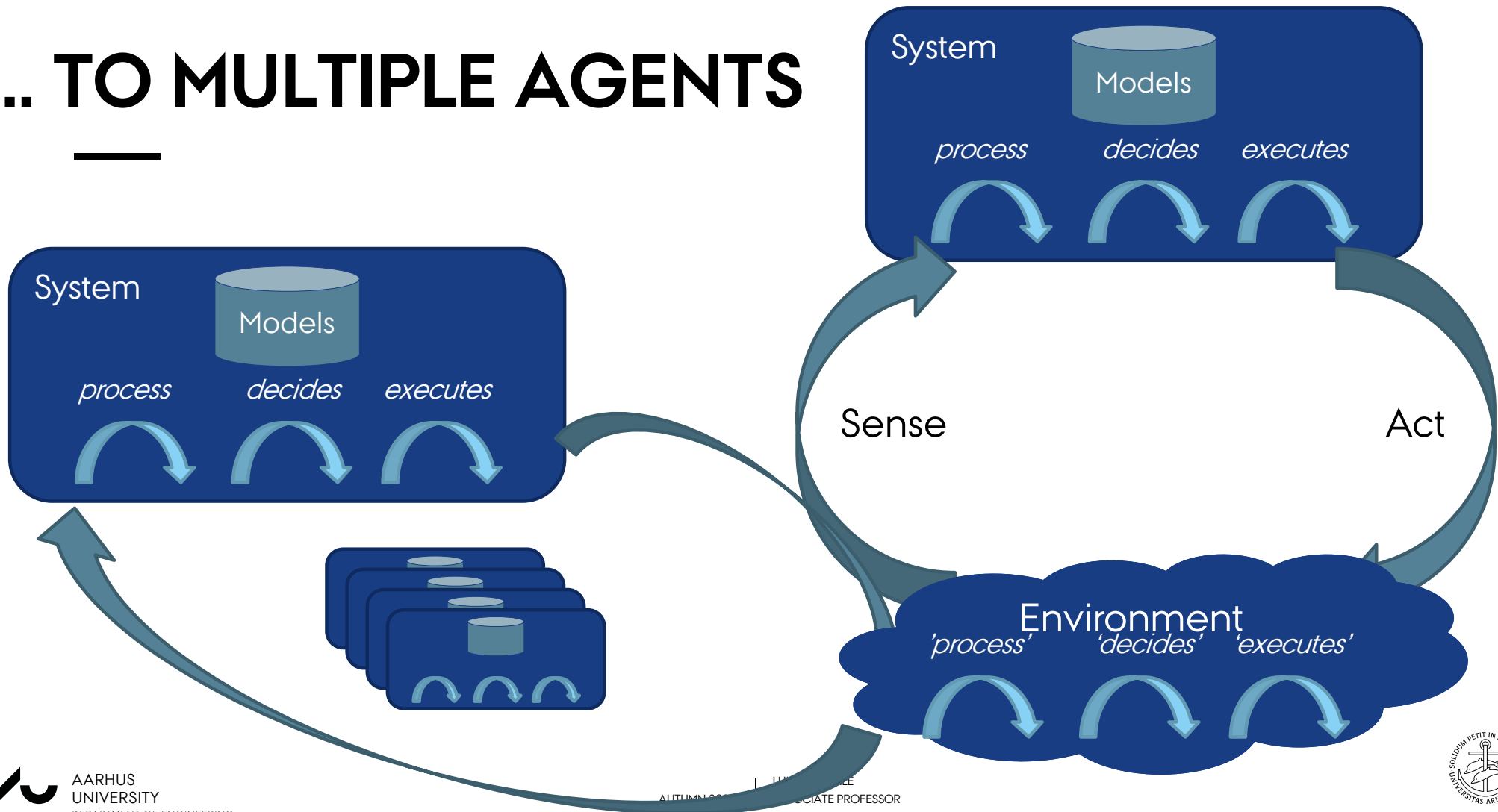
Mirgita Frasheri

With slides from Prof. Christian Bettstetter (Klagenfurt University)

FROM SINGLE AGENTS...



... TO MULTIPLE AGENTS



RULES GOVERNS HOW WE INTERACT



CHARACTERISTICS

Locality – only information that is locally available

Autonomy – every agent is in control of own actions, resources, goals, ...

Decentralised – no dedicated leader

Self-organization – coordination and competition emerges without central control

Communication – ability to share local information

- **Direct:** send messages directly to other agents
- **Global:** send messages to everyone
- **Environmental:** leave message in the local environment



POSSIBLE COORDINATION (MORE IN 2 WEEKS)

Consensus and Negotiation – agents negotiate/agree resources and tasks

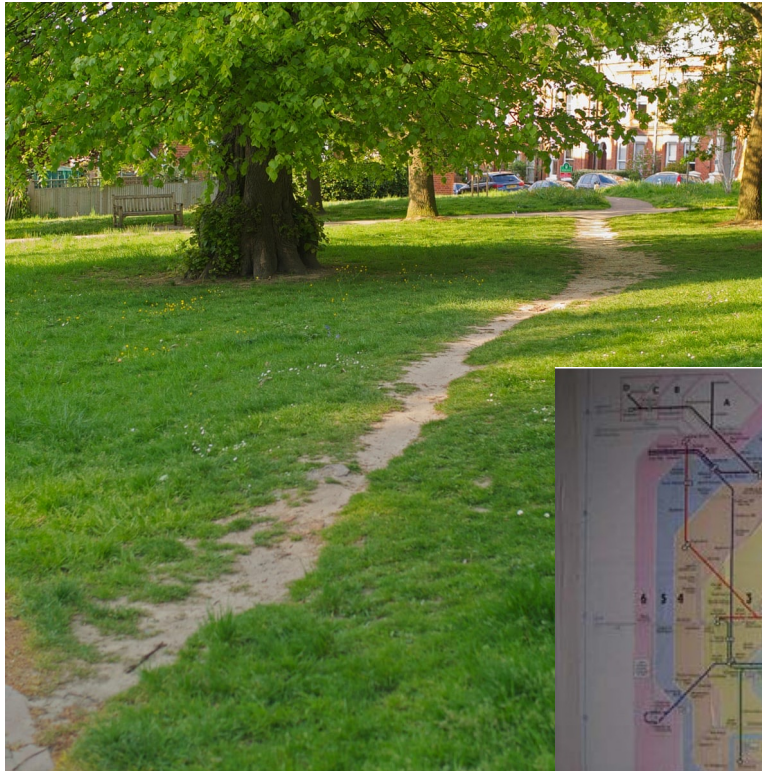
- Auctions
- Bargaining
- Voting

Coalition formation – agents with same/similar goals work together

Leader election – select a leader temporarily or from the current cohort

STIGERGY AND EMERGENCE

—
stigma (mark, sign) +
ergon (work, action)



1959 by Pierre-Paul Grassé



EMERGENT BEHAVIOUR

Nature:

- Flocks of birds
- Ants & Termites
- Fish
- Sand
- ...

Humans:

- Stock market
- Architecture
- Traffic



CONVEY'S GAME OF LIFE

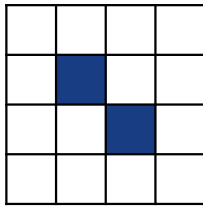
Grid based world with cells either *dead* or *alive* invented by John Conway

Allows for emergent behaviour from initial state with three simple rules:

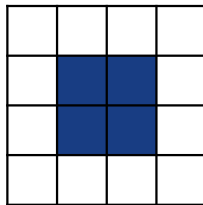
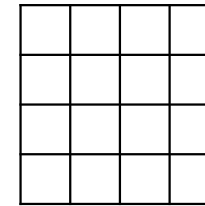
- An *alive* cell with two or three live cells survives
- A *dead* cell with exactly three living cells becomes alive
- All other cells die (or remain dead) in the next iteration

Martin, G. (1970). Mathematical Games: The fantastic combinations of John Conway's new solitaire game "Life". *Scientific American*, 223, 120-123.

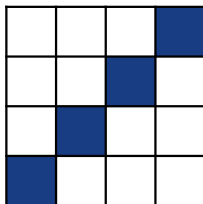
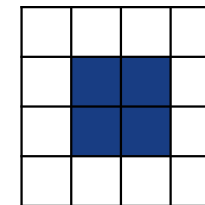
CONVEY'S GAME OF LIFE



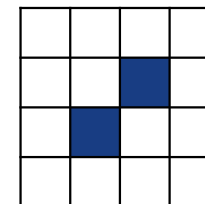
Death



Survive

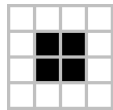


Reproduce

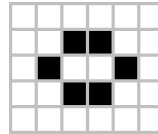


SEVERAL STRUCTURE IN GAME OF LIFE

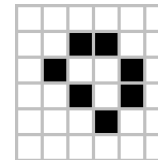
Static Structures (Still lifes)



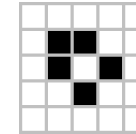
Block



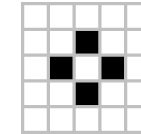
Bee hive



Loaf



Boat

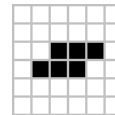


Tub

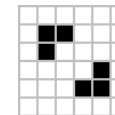
Oscillators



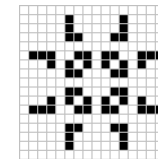
Blinker



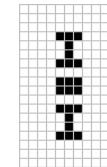
Toad



Beacon

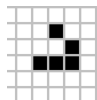


Pulsar

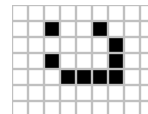


Penta-decathlon

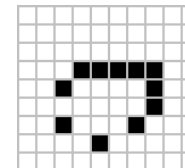
Moving (Space Ships)



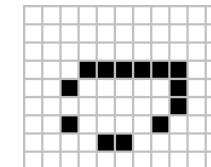
Glider



Light
Spaceship



Middle-weight
Spaceship



Heavy-weight
Spaceship

CONVEY'S GAME OF LIFE

- <https://playgameoflife.com/>

SYNCHRONISATION

- Firefly swarms
- Hands clapping
- Menstrual cycles
- Heart cells
- Brain activity
- Bride vibrations
- ...



“Synchronous Fireflies” in Smoky Mountains National Park

© Radim Schreiber. fireflyexperience.org

FIREFLY SYNCHRONISATION



PULSE COUPLED OSCILLATORS



SYNCHRONISATION

—
Greek:

Syn (= together) +
chronos (= time)



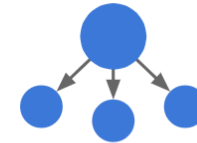
SYNCHRONISATION

Definitions

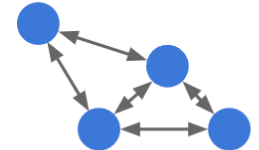
- Synchronous: events occur at the same time (in sync)
- Synchronisation: process leading to synchrononous events

Classification and performance

- Various types and classes of synchronisation
- Performance metrics: convergence probability, time to synchrony, precision, scalability, robustness, etc.



Master-slave sync
(monarchy)



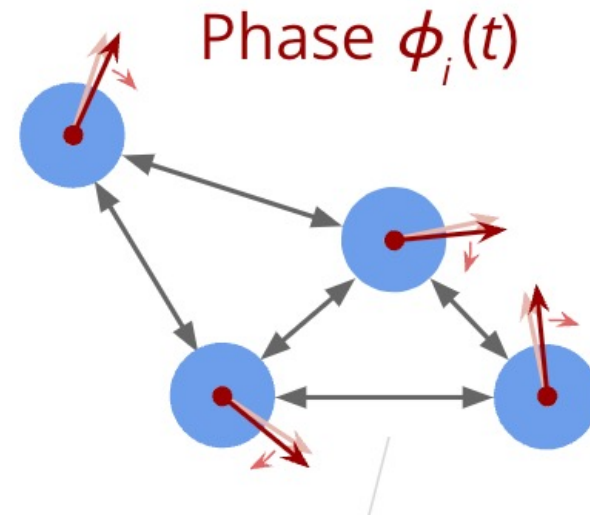
Mutual sync
(base democracy)

↓
Self-organization

Slide by Christian Bettstetter (Klagenfurt University)

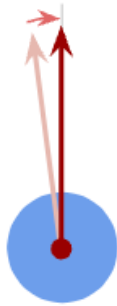
MUTUAL SYNC: COUPLED OSCILLATORS

- Each oscillator i has a phase ϕ_i that evolves over time t .
- The phase evolution is influenced via **coupling** with **other oscillators**.
- **Coupling** involves
 - Exchanging of state information
 - Changing its own **phase** as a reaction, using **phase update function**
- **Coupling** can be time-continuous or time-discrete. Discrete coupling is often done via the exchange of pulses.



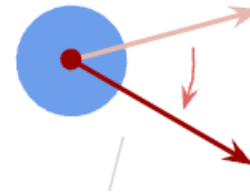
Slide by Christian Bettstetter (Klagenfurt University)

PULSE-COUPLED OSCILLATORS (PCOs)



The **oscillator** emits a **pulse** signal.

The **phase** of an **oscillator** i reaches a **threshold** $\phi = 0$.



Each **oscillator** receiving the **pulse** performs a **phase jump** $\Delta\phi$:

$$\phi_j(t^+) = \phi_j(t) \Delta\phi(\phi_j(t))$$

PULSE-COUPLED OSCILLATORS

- **How far do the phases jump?**

The **phase jumps** are defined by a **phase update function**.
Jumps can be *forward* (excitatory) or *backward* (inhibitory).

- **What data do oscillators exchange?**

The **pulse signals** contain *no data*.
All state information is encoded in the timing of the pulses.

- **Is synchrony always achieved?**

There are proofs that PCOs converge to synchrony, but they typically rely on specific *assumptions* (like all-to-all, coupling, perfect links, and identical oscillators).

FROM AGENTS TO MULTI-AGENT SYSTEMS

Cooperation: working together towards (individual) goals without intention



Collaboration: working with intention towards a *common* goal



Competition: working selfishly towards own goals without consideration of others



ANT FORAGING

- Ants use indirect communication of *pheromones*
- Pheromones are left by ants with some probability when encountering pheromones in the environment
- Pheromones evaporate over time (disappear)
- The stronger pheromones are, the more likely other ants will leave pheromones

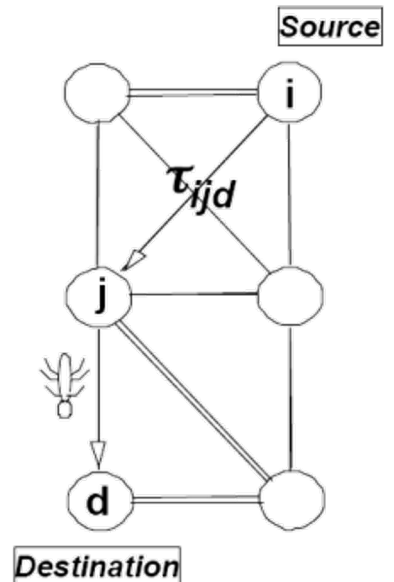
ARTIFICIAL ANTS

- Live in a discrete world
- Deposit pheromone in a problem dependent way
- Can have extra capabilities: Local search, lookahead, backtracking
- Exploit an internal state (memory)
- Deposit an amount of pheromone function of the solution quality
- Can use local heuristic

ANT COLONY OPTIMISATION

- Ants are launched at regular instants from each node to randomly chosen destinations
- Ants build their paths probabilistically with a probability function of:
 1. artificial pheromone values, and
 2. heuristic values
- Ants memorize visited nodes and costs incurred
- Once reached their destination nodes, ants retrace their paths backwards, and update the pheromone trails

The pheromone trail is the stigmergic variable



FORMALLY

Pheromone deposition

$$\tau_{ijd}(t+1) \leftarrow \tau_{ijd}(t) + \Delta_{ijd}^k(t)$$

Heuristic state updating rule

$$p_{ijd}^k(t) = f(\tau_{ijd}(t), \eta(t))$$

Probabilistic state updating rule

$$p_{ijd}^k(t) = \frac{[\tau_{ijd}(t)]^\alpha \cdot [\eta_{ijd}(t)]^\beta}{\sum_{h \in J_i^k} [\tau_{ihd}(t)]^\alpha \cdot [\eta_{ihd}(t)]^\beta}$$

Evaporation rule

$$\tau_{ijd}(t+1) \leftarrow (1 - \rho) \cdot \tau_{ijd}(t)$$

τ_{ijd} is the amount of pheromone trail on edge (i, j, d)

η_{ijd} is the heuristic evaluation of link (i, j, d)

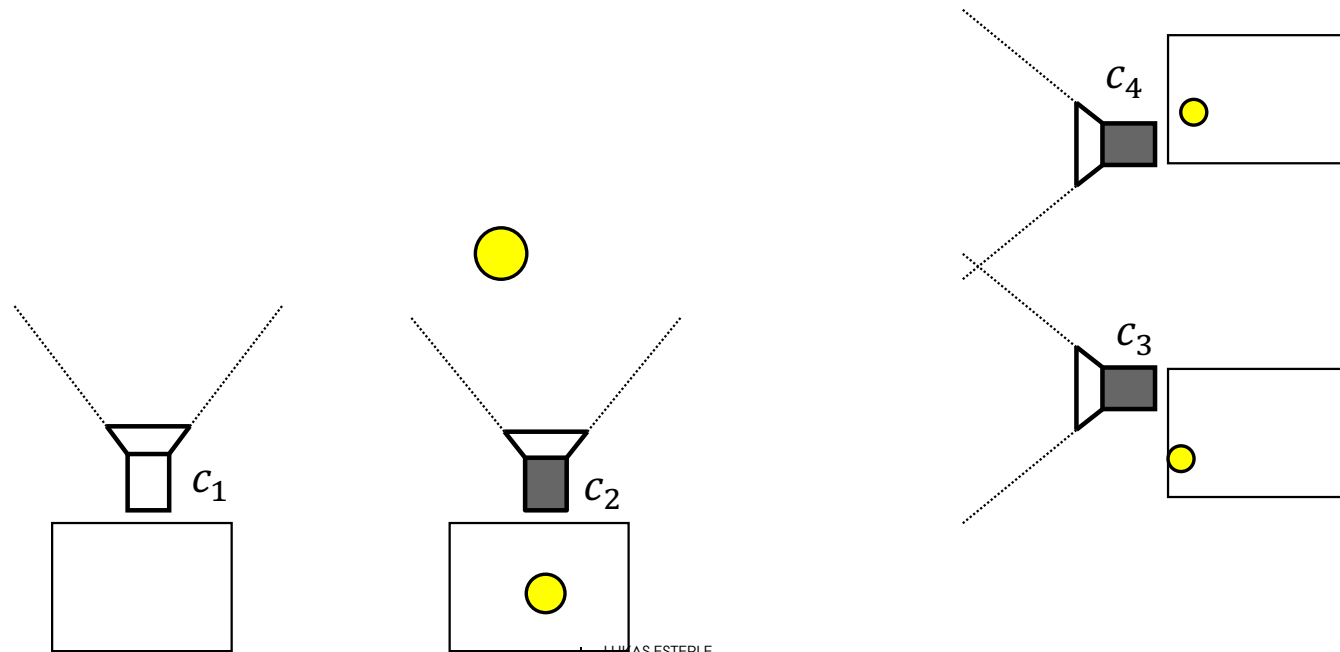
J_i^k is the set of feasible nodes ant k positioned on node i can move to

WHY DOES IT WORK?

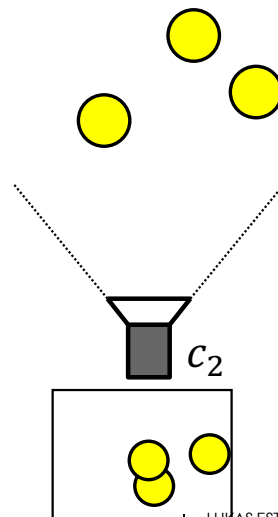
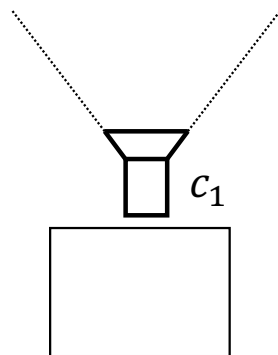
Three important components:

- **TIME:** a shorter path receives pheromone quicker (this is often called: "differential length effect")
- **QUALITY:** a shorter path receives more pheromone
- **COMBINATORICS:** a shorter path receives pheromone more frequently because it is likely to have a lower number of decision points

DISTRIBUTED TRACKING IN CAMERAS

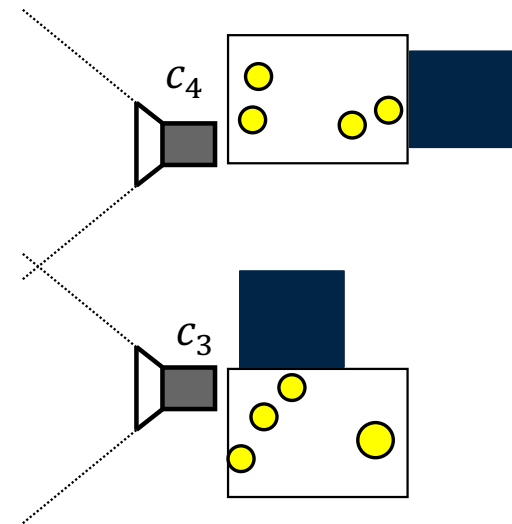


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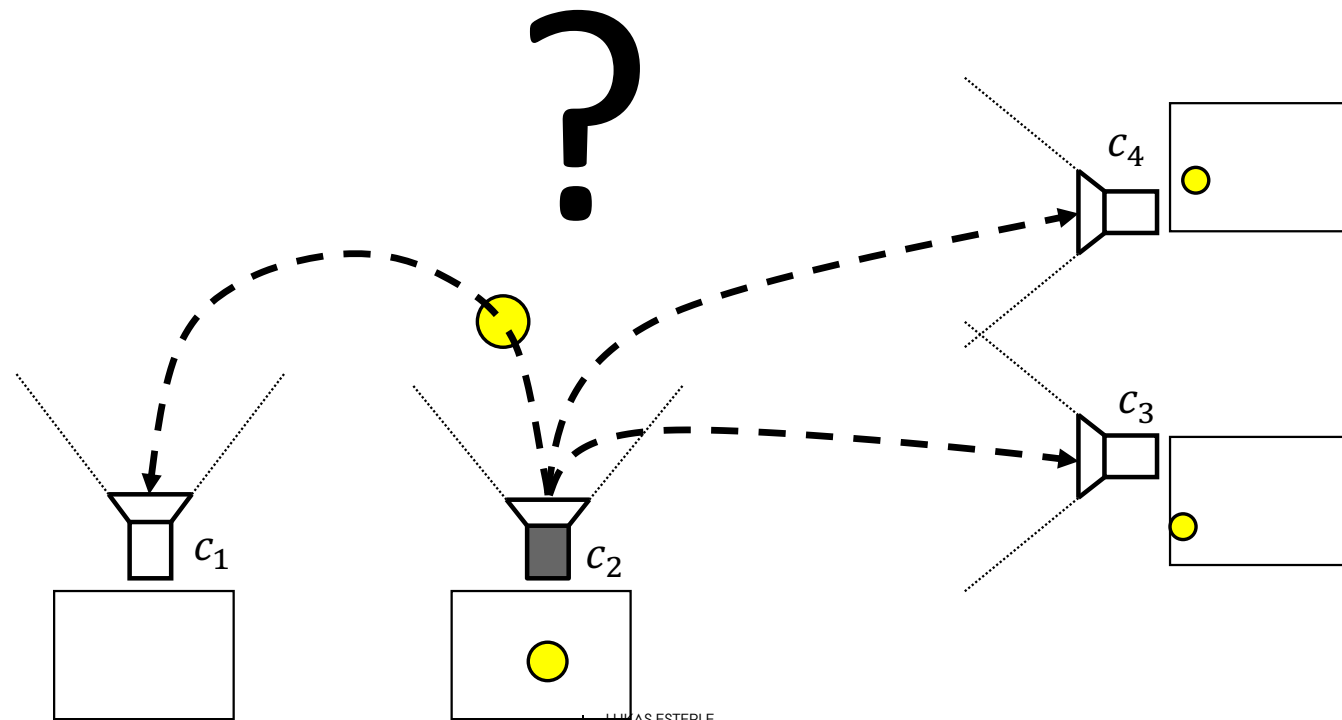


AUTUMN 2024

LUKAS ESTERLE
ASSOCIATE PROFESSOR



—



CREATING THE VISION GRAPH

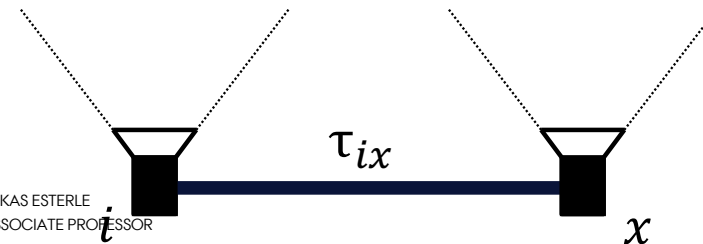
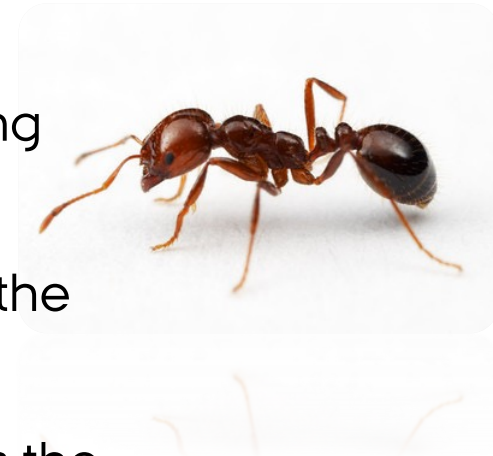
Artificial pheromones inspired by the ant foraging process.

The selling camera i creates a link τ_{ix} to the buying camera x for future trading purpose.

Local vision graph for each camera representing the local neighbourhood.

Multiple trades with the same camera strengthens the link between cameras.

Introduces a “forgetting” factor.



EXPLOITING THE VISION GRAPH

In addition to advertise auctions to all cameras, we came up with two communication policies:

- STEP

Probability to communicate with another node is based on the **strength of the link**. All others have only a **small probability** to get the invitation as well.

$$P_{STEP(i,x)} = \begin{cases} 1 & \text{if } \tau_{ix} > \varepsilon \\ \eta & \text{otherwise} \end{cases}$$

- SMOOTH

Probability to communicate with another node is based on the strength of the **link relative to strongest link**.

$$P_{SMOOTH(i,x)} = \frac{1 + \tau_{ix}}{1 + \tau_{im}}$$
$$m = \operatorname{argmax}_y \tau_{iy}, \forall y$$

y

CRAIG REYNOLD'S BOIDS

Boids simulate flocking behaviour of birds

Relies on 3 simple rules

Used to animate flocks of animals in movies (e.g. Batman Returns)

Initially designed with all boids being equal

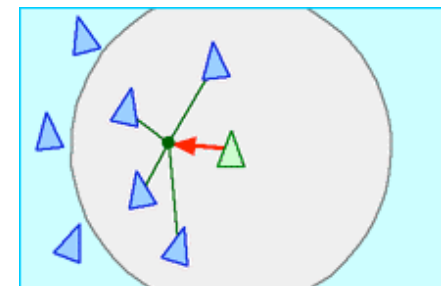
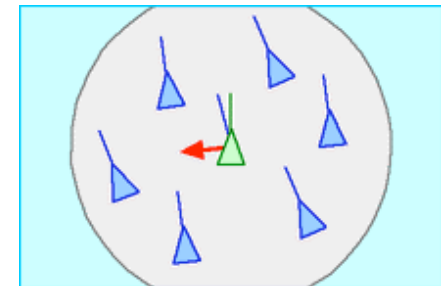
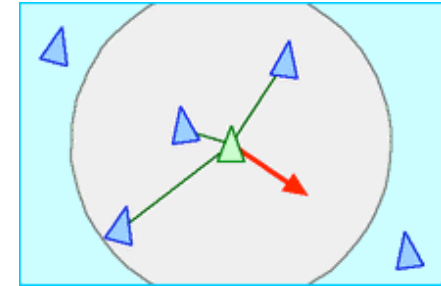
Later leadership and more complex rules have been added.



CRAIG REYNOLD'S BOIDS

Simple rules to achieve flocking behaviour:

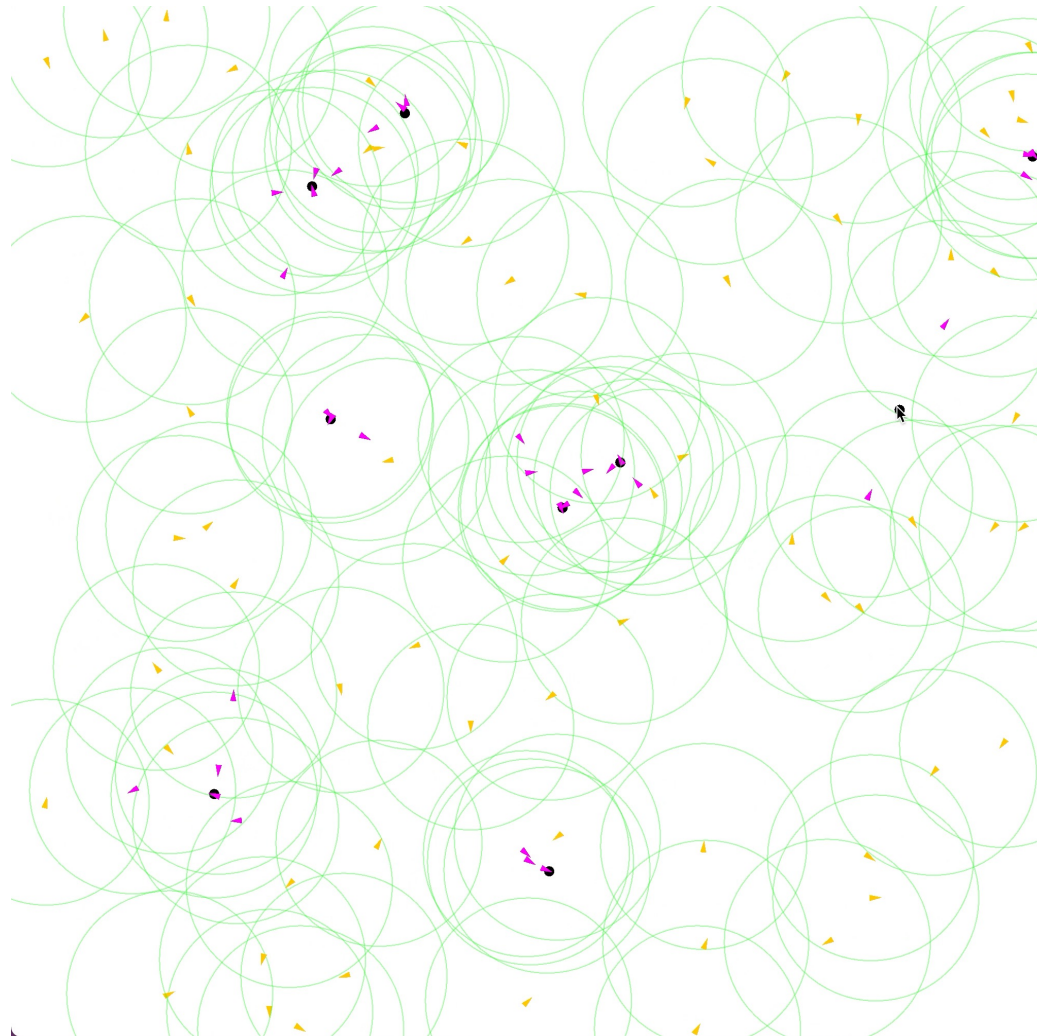
- **separation:** steer to avoid crowding
- **alignment:** steer towards the average heading
- **cohesion:** steer to move towards the average position (center of mass) of local flockmates



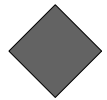
CRAIG REYNOLD'S BOIDS

<http://www.harmendeweerd.nl/boids/>

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ONLINE MULTI-OBJECT K-ASSIGNMENT



Target

- Appear randomly
- Move around



Observer

- Can sense environment
- Can talk to other observer
- Have a set of tracker

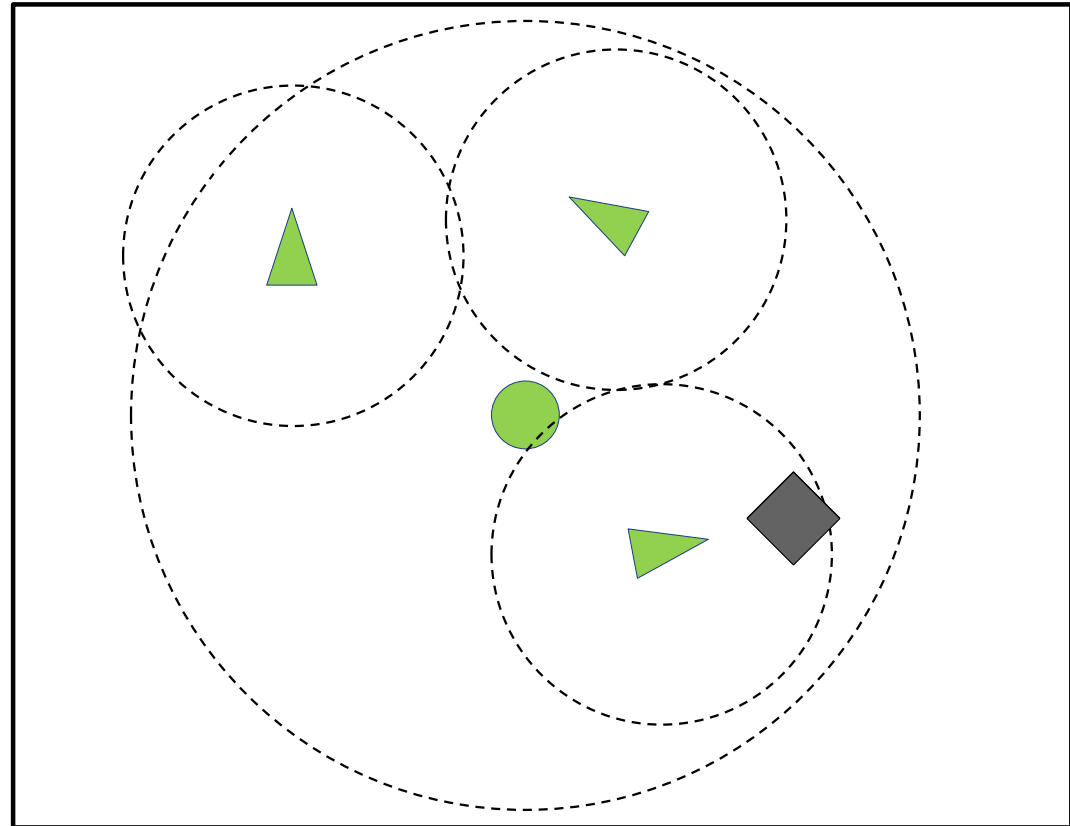


Follower

- Can detect targets
- Follow assigned targets
- Communicate with other tracker

Goal

- Assign exactly k tracker to each target
- Maximise the number of detected targets



THREE COORDINATION STRATEGIES

Autonomous

- Only *followers*, no *observer*
- Each *follower* operates completely autonomous - no central control

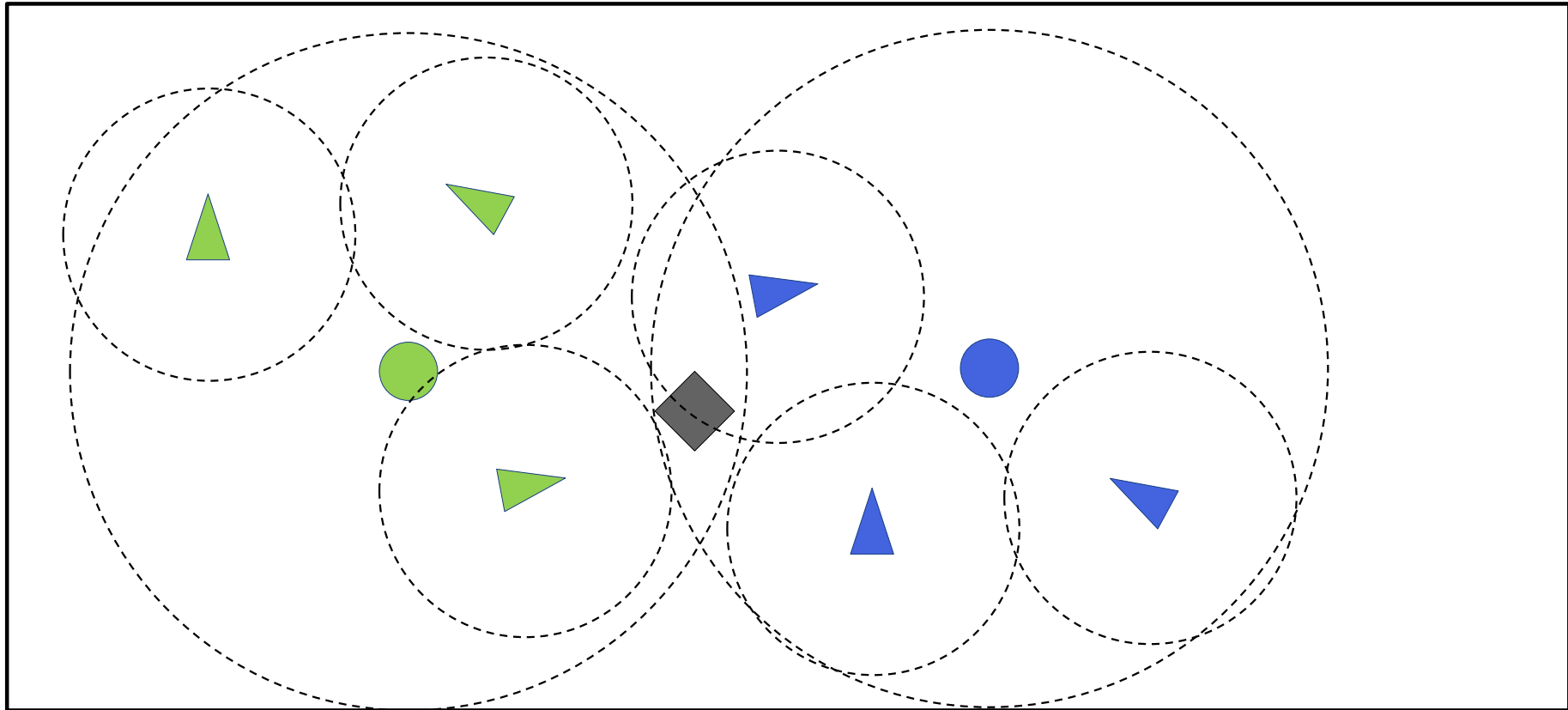
Tightly supervised

- *Followers* are assigned only by respective *observer* to each object
- *Observer* coordinate among each other

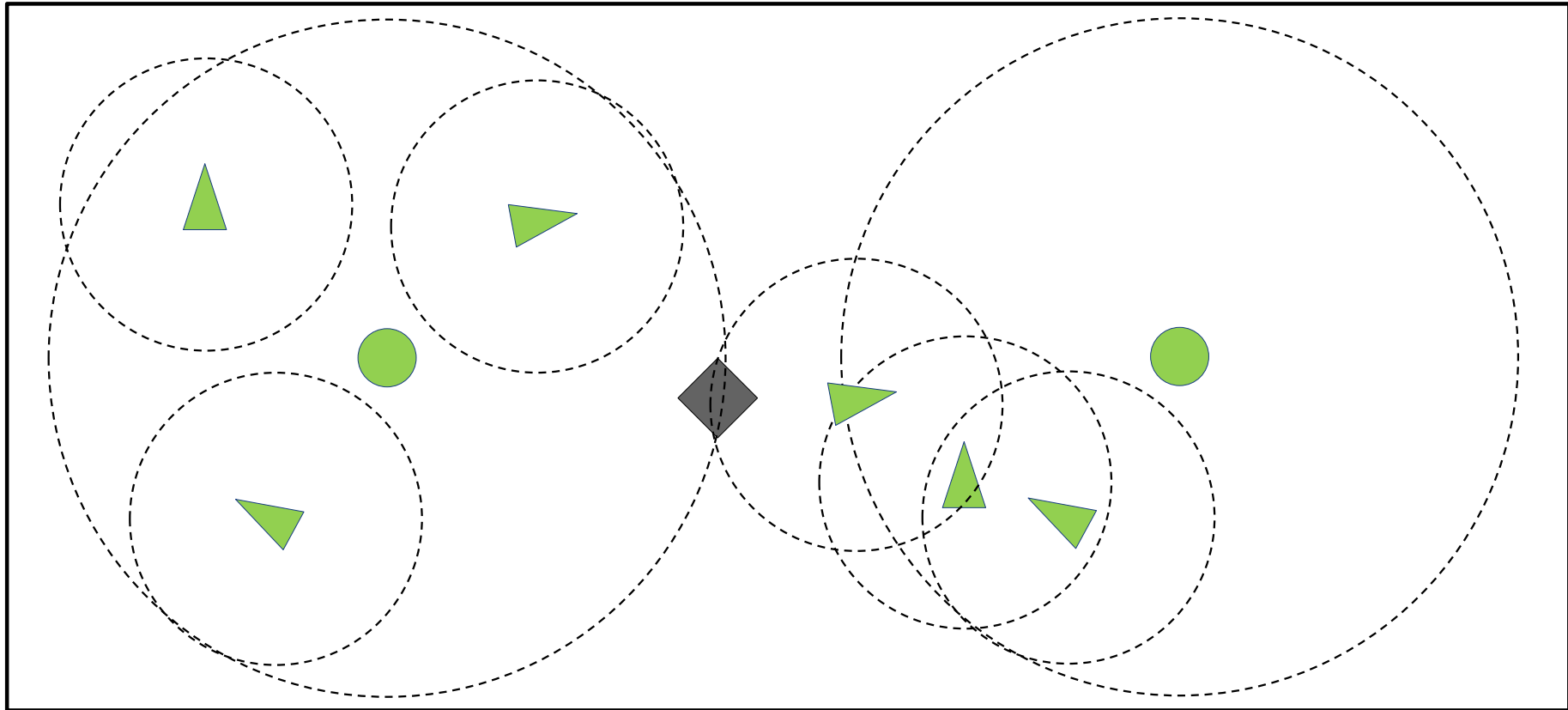
Loosely supervised

- *Followers* are assigned by *observer*
- *Follower* can re-assign based on local information

TIGHT SUPERVISION - PROBLEM



AUTONOMY AND LOOSE SUPERVISED



CONCLUSION

Observer agents with larger sensory and communication ranges help cover larger areas and contribute to the detection of agents

Autonomous tracking agents (PA and LSC) help avoid overprovisioning

All agents were designed to collaborate

SUMMARY

Everything can be an agent

Rules govern how we interact with others

Stigmery evolves from the interaction in the environment

Emergence evolves from the interaction with each other (Game of life)

Nature as inspiration to solve problems (ants, fireflies, and birds)